



PRN : 25070122073

NAME: DHARMIT KUNAL SHAH

BATCH: CSE A-2

Subject: Programming with Java

ASSIGNMENT No : 15

TITLE: Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

```
package Exp15;

class InvalidAgeException extends Exception {
    public InvalidAgeException(String m) {
        super(m);
    }
}

public class Main1 {
    public static void validate(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException(m: "Age must be 18 or above");
        }
        System.out.println("Valid age : " + age);
    }

    Run | Debug
    public static void main (String[] args) {
        try {
            validate(age: 12);
        } catch (InvalidAgeException e) {
            System.out.println("Caught Exception : " + e.getMessage());
        }
    }
}
```

```
package Exp15;

public class Main2 {
    Run | Debug
    public static void main (String[] args) {
        try {
            int[] n = new int[5];
            int divisor = 0;

            for (int i = 0; i < n.length; i++) {
                int res = n[i] / divisor;
                System.out.println(res);
            }
        } catch (ArithmeticException e) {
            throw new RuntimeException(message: "Error : Division by zero occurred", e);
        }
    }
}
```

```

PS D:\Java-Questions\Exp15> java Main1.java
Caught Exception : Age must be 18 or above
PS D:\Java-Questions\Exp15> java Main2.java
Exception in thread "main" java.lang.RuntimeException: Error : Division by zero occurred
    at Exp15.Main2.main(Main2.java:14)
Caused by: java.lang.ArithmeticException: / by zero
    at Exp15.Main2.main(Main2.java:10)
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:104)
    at java.base/java.lang.reflect.Method.invoke(Method.java:565)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.execute(SourceLauncher.java:264)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.run(SourceLauncher.java:138)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.main(SourceLauncher.java:76)
PS D:\Java-Questions\Exp15>

```

Exercise

1. Create a Login program that throws an exception for invalid password and uses finally block.

```

// Create a Login program that throws an exception for invalid password and uses finally block
package Exp15.Exercise;

import java.util.Scanner;

public class Exp1 {
    Run | Debug
    public static void main (String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            String correctPassword = "admin123";
            try {
                System.out.print(s: "Enter Password : ");
                String password = in.nextLine();

                if (!password.equals(correctPassword)) {
                    throw new Exception(message: "Invalid Password!");
                }

                System.out.println(x: "Login Successful.");
            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            } finally {
                System.out.println(x: "Login process completed.");
            }
        }
    }
}

```

```

PS D:\Java-Questions\Exp15\Exercise> java Exp1.java
Enter Password : admin123
Login Successful.
Login process completed.

```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```
// Create an ATM PIN Verification program that throws an exception for an
// invalid PIN entered by the user and uses a finally block to display a message indicating th
package Exp15.Exercise;

import java.util.Scanner;

public class Exp2 {
    Run | Debug
    public static void main (String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            try {
                String pass = "12345";

                System.out.print(s: "Enter ATM PIN : ");
                String pin = in.next();

                if (!pin.equals(pass)) {
                    throw new Exception(message: "Incorrect ATM PIN!");
                }

                System.out.println(x: "Logging in...");
                System.out.println(x: "ATM Check in Verified!!!");
            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            } finally {
                System.out.println(x: "ATM Verification Completed!");
            }
        }
    }
}
```

```
PS D:\Java-Questions\Exp15\Exercise> java Exp2.java
Enter ATM PIN : 12345
Logging in...
ATM Check in Verified!!!
ATM Verification Completed!
PS D:\Java-Questions\Exp15\Exercise> █
```